DATABASE MANAGEMENT SYSTEM

SQL QUERIES

ASSIGNMENT- 3

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Write an SQL query to display all student records from the Student table.

```sql
CREATE TABLE Std1(ID INT, Name VARCHAR(20), Dept VARCHAR(20), Marks INT, City VARCHAR(20));
INSERT INTO Std1 VALUES (201, 'SAM ', 'CSE', 85, 'Chennai');
INSERT INTO Std1 VALUES (202, 'HARAN', 'ECE', 78, 'Coimbatore');
INSERT INTO Std1 VALUES (203, 'KAVIN ', 'CSE', 92, NULL);
INSERT INTO Std1 VALUES (204, 'GURU', 'IT', 74, 'Chennai');
INSERT INTO Std1 VALUES (205, 'AAKASH', 'ECE', 88, 'Madurai');
INSERT INTO Std1 VALUES (201, 'PRIYA ', 'CSE', 85, 'Chennai');
SELECT * FROM Std1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
202	HARAN	ECE	78	Coimbatore
203	KAVIN	CSE	92	NULL
204	GURU	IT	74	Chennai
205	AAKASH	ECE	88	Madurai
201	PRIYA	CSE	85	Chennai

Write a query to display the Name and Marks of all students.

```sql
SELECT Name, Marks FROM Std1;
```

Result Grid | Filter Rows:

Name	Marks
SAM	85
HARAN	78
KAVIN	92
GURU	74
AAKASH	88
PRIYA	85

Write a query to display students who scored more than 80 marks.

```
1 •  SELECT * FROM Std1 WHERE Marks>80;
```

Result Grid | Filter Rows: | Exp

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
203	KAVIN	CSE	92	NULL
205	AAKASH	ECE	88	Madurai
201	PRIYA	CSE	85	Chennai

Write a query to display students belonging to the CSE department.

```
1 •  SELECT * FROM Std1 WHERE Dept='CSE';
```

Result Grid | Filter Rows: | Export:

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
203	KAVIN	CSE	92	NULL
201	PRIYA	CSE	85	Chennai

Write a query to display students whose city is Chennai.

```
1 •  SELECT * FROM Std1 WHERE City='Chennai';
```

Result Grid | Filter Rows: | Export:

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
204	GURU	IT	74	Chennai
201	PRIYA	CSE	85	Chennai

Write a query to display students whose marks are between 70 and 90.

```
1 •  SELECT * FROM Std1 WHERE Marks BETWEEN 70 AND 90;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
202	HARAN	ECE	78	Coimbatore
204	GURU	IT	74	Chennai
205	AAKASH	ECE	88	Madurai
201	PRIYA	CSE	85	Chennai

Write a query to display student names that end with the letter 'a' using the LIKE operator.

```
1 •  SELECT Name FROM Std1 WHERE Name LIKE '%a';
```

Result Grid | Filter Rows: | Export: | W

Name
KAVIYA

Write a query to display distinct departments from the Student table.

```
1  SELECT DISTINCT Dept FROM Std1;
```

Result Grid | Filter Rows: | E

Dept
CSE
ECE
IT

Write a query to display students whose City value is NULL.

```
SELECT * from Std1 where City is NULL;
```

Result Grid | Filter Rows: | Export: | W

ID	Name	Dept	Marks	City
203	KAVIN	CSE	92	NULL

Write a query to find the highest marks obtained by a student using an aggregate function.

```
SELECT MAX(Marks) AS Highest_Marks FROM Std1;
```

Result Grid | Filter Rows: | Export: | Wr

Highest_Marks
92

Write a query to find the average marks of students in the ECE department.

```
SELECT AVG(Marks) AS Average_Marks from Std1 where Dept='ECE';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Average_Marks
83.0000

Write a query to count the number of students in each department using GROUP BY.

```
SELECT Dept, COUNT(*) AS Number_of_students FROM Std1 GROUP BY Dept;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Dept	Number_of_students
CSE	3
ECE	2
IT	1

Write a query to display students whose marks are greater than the average marks of all students (nested subquery).

```
SELECT * FROM Std1 where Marks> (SELECT AVG(Marks) from Std1);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ID	Name	Dept	Marks	City
201	SAM	CSE	85	Chennai
203	KAVIN	CSE	92	NULL
205	AAKASH	ECE	88	Madurai
201	PRIYA	CSE	85	Chennai

Write an SQL statement to insert a new student record into the Student table.

```
INSERT INTO Std1 VALUES (206, 'KAVIYA', 'CIV', 82, 'KOVAI');
SELECT * FROM Std1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ID	Name	Dept	Marks	City
203	KAVIN	CSE	92	NULL
204	GURU	IT	74	Chennai
205	AAKASH	ECE	88	Madurai
201	PRIYA	CSE	85	Chennai
206	KAVIYA	CIV	82	KOVAI

Write an SQL query to delete the student with StudentID = 205.

```sql
SET SQL_SAFE_UPDATES=0;
DELETE FROM Std1 WHERE ID= 201;
SELECT* FROM Std1;
```

Result Grid | Filter Rows: | Export:

ID	Name	Dept	Marks	City
202	HARAN	ECE	78	Coimbatore
203	KAVIN	CSE	92	NULL
204	GURU	IT	74	Chennai
205	AAKASH	ECE	88	Madurai
206	KAVIYA	CIV	82	KOVAI